2017-Ph-H1-Q25

Section: Electricity
Topic: Semiconductors, Sources, Internal Resistance

> An LED has a **forward voltage** of 2.0 V.
> It is connected to a 9.0 V supply through a resistor.
> What is the **minimum resistance** to limit the current to 10 mA?

🔍 Step-by-step:

1. Voltage across resistor:

$$V_R = 9.0 - 2.0 = 7.0 \text{ V}$$

2. Use Ohm's Law:

$$R = \frac{V_R}{I} = \frac{7.0}{0.010} = 700\ \Omega$$

✅ Final Answer: D — 700 Ω

🧠 Revision Tips:

- LEDs have a **threshold voltage** — no current flows below it
- Always subtract LED forward voltage from supply voltage